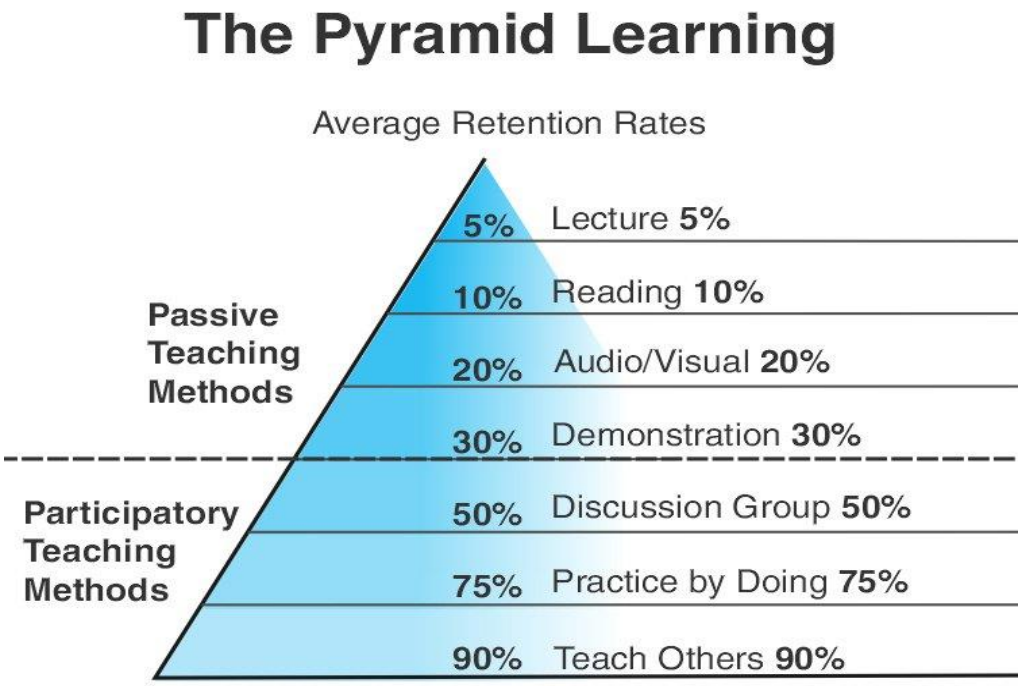
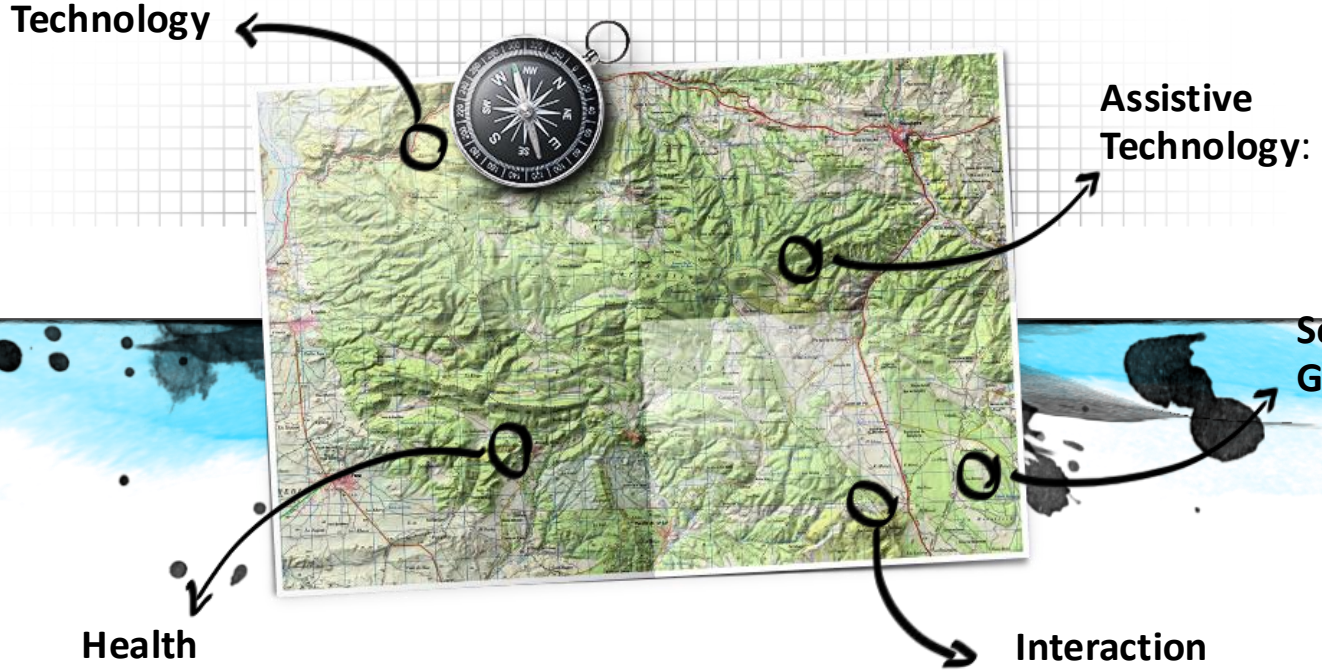

isep
instituto
superior de
engenharia do
porto

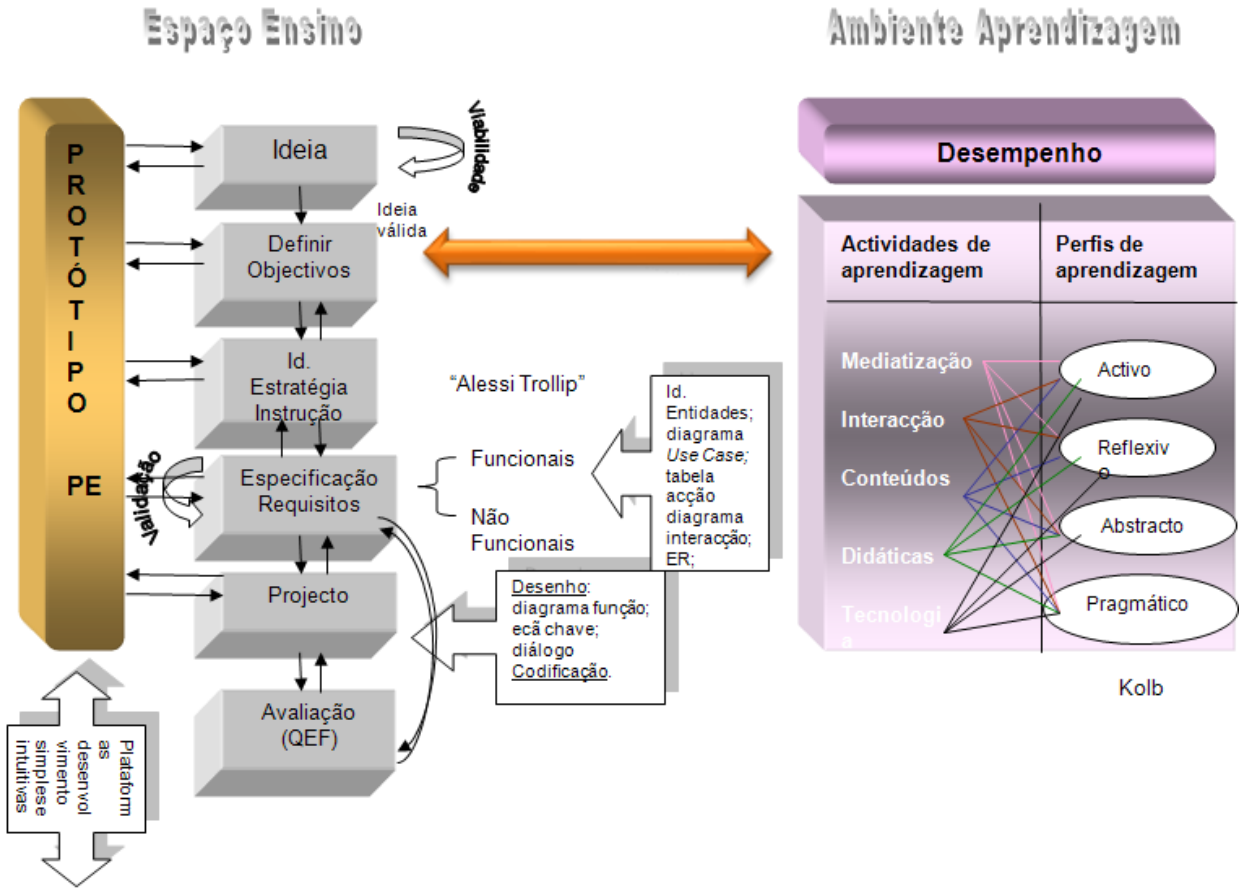
Benefits of active learning



HEALTH
ENGENEERING
EDUCATION
BUSINESS
ART
PSYCHOLOGY
SCIENCE
SPORT

Research areas





Techno-Didactical Extension for Instruction/Learning Based on Computer

Object-based and result-oriented

Quantitative Evaluation Framework

QEF

QEF

Quantitative Evaluation Framework:

Quantitative evaluation of the quality of educational software.

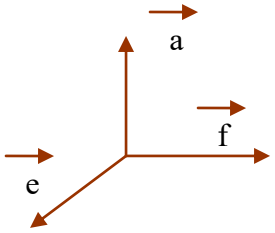
It is based on the principles of Software Engineering (Pressman, 2001) and quality concepts (Bates, 2000).

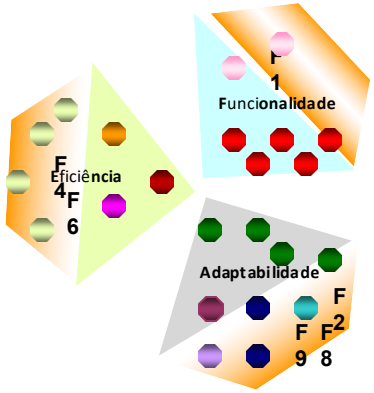
Reference Standards:
ISO 9126
SCORM

QEF

Quantitative evaluation of the quality of educational software.

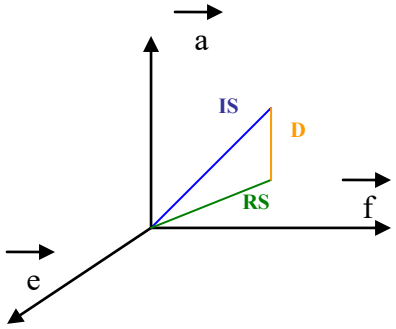
Space in three dimensions: Functionality, Adaptability and Efficiency





Each dimension consists of a set of Factors and each Factor of a set and requirements.

QEF



The quality of the System is inversely proportional to the distance between the Ideal System (IS) and the Real System (RS).
If D=0 then Q=1
If D=maxim, D= max
So Q=0

Each dimension consists of a set of Factors and each Factor of a set and requirements.

Objective: To obtain the level of performance of the system at any stage of development

QEF

Quantitative Evaluation Framework:

- Quantitative evaluation of the quality of educational software.

The quality q, of a given educational software, is defined in our three-dimensional quality space, Q, and represents the measure, in percentage values, in relation to the hypothetical ideal educational software, I, represented by the coordinates (1,1,1).

QEF

Quantitative Evaluation Framework:

- Quantitative evaluation of the quality of educational software.

For each system to be validated, we must define the relative importance of each factor for each of the dimensions:

$$\text{Dimensão}_i = \sum_n (p_n \times factor_n) \quad \sum_n (p_n) = 1 \qquad p_n \in [0,1]$$

N is the number of factors relevant to the dimension in question.

QEF

Quantitative Evaluation Framework:

- Quantitative evaluation of the quality of educational software.

Each factor is assessed as follows:

$$\text{Factor } n = \frac{1}{\sum_m pr_m} \times \sum_m (pr_m \times pc_m) \quad p_n \in [0,1]$$

Since m is the number of criteria relevant to the factor in question, the weight of criterion m and pcm is the percentage of compliance with criterion m.

QEF

Quantitative Evaluation Framework:

- Quantitative evaluation of the quality of educational software.

The overall deviation of the system (Euclidean distance) from the ideal system will then be given by:

$$D = \sqrt{\sum_j \left(1 - \frac{Dim_j}{100}\right)^2}$$

And the Quality of the system will be:

$$Q = 1 - \frac{D}{\sqrt{n}}, Q \in [0,1] \quad \text{ou} \quad q = \left(1 - \frac{D}{\sqrt{n}}\right) * 100 \; q \in [0,100]$$

The quality of a system is a measure of the distance between the designed or ideal system and the actual or actual produced system

SW Product Quality – ISO/IEC 9126

Functionality	product-specific functions and properties, meet the user's needs
Reliability	The software product is able to maintain its level of performance, over time, under the established conditions.
Usability	effort required to use the system, based on a set of user implications and conditions.
Efficiency	The features and times involved are commensurate with the level of performance required by the software.
Maintenance	refers to the effort required to make specific changes to the software product.
Portability	ease of software being transferred from one environment to another

Example1: Identification of Factors for Evaluation of SWE

Id	Factor
F1	Ease of use and installation
F2	Versatility
F3	Quality around the audiovisual
F4	Aesthetic and technical elements
F5	Content quality
F6	Navigation and interaction
F7	Mandatory and use of advanced technology
F8	Pedagogical aspects
F9	Potentialities of two teaching resources
F10	Fostering initiative and self-learning
F11	Cognitive exertions that require your activities

Example 1: Identification of Factors for SWE Evaluation

Dimension	Id	Factor	Id	Requirement
Functionality	F1	**Ease of use and installation**	1	The user can use it immediately without having to carry out an exhaustive reading of the manuals
			2	There is an online help system to solve any doubts that may arise
	F5	**Content quality**	8	The information is well-structured, adequately differentiating objectives, opinions, and fictional elements
			9	The texts have no spelling errors and the sentence construction is correct
			10	The contents and messages are not negative or biased, and there is no discrimination based on social class, beliefs, religion, etc.
			11	The contents are related to situations and problems of the user's interest
			12	There are examples, simulations, and graphics

Example 1: Identification of Factors for SWE Evaluation

Dimension	Id	Factor	Id	Requirement
Adaptability	F2	**Versatility**	3	It can be easily integrated with other didactic means
			4	Allows modification of some parameters (level of difficulty, number of simultaneous users, language, …)
			5	Includes an evaluation and monitoring system (errors, predefined itineraries to solve problems)
	F8	**Pedagogical aspects**	19	Allows activities to awaken and maintain users' curiosity and interest in the subject matter of the content, without causing anxiety
	F9	**Potential of didactic resources**	20	Proposes various types of activities that allow different ways of using knowledge

Example 1: Identification of Factors for SWE Evaluation

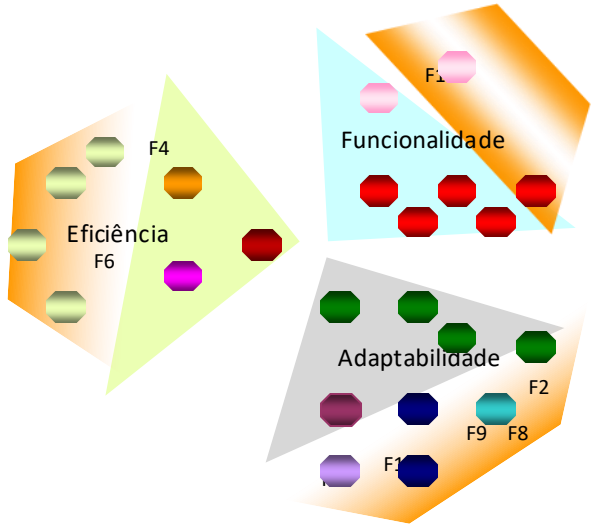
Dimension	Id	Factor	Id	Requirement
Eficiency	F4	**Aesthetic elements and techniques**	7	Titles, menus, video, sound, photographs, metaphors, composition, color.
	F6	**Navigation and interaction**	13	The educational software has a good program structure that allows proper access to content and activities.
			14	The communication speed between the program and the user (animation, content presentation, data reading, …) is adequate.
			15	The program runs reliably and has no execution errors.
			16	The navigation system is transparent, allowing the user to control actions.
	F7	**Mandatory use of advanced technology**	17	The program was developed around originality, clearly differentiated from other didactic materials, and makes use of multimedia technologies.

Identifying Quality Space Dimensions

Functionality:
Ease of use and installation (F1)
Quality of content. (F5)

Adaptability:
Versatility (F2)
Pedagogical aspects (F8)
Potential of didactic resources (F9)
Fostering initiative and self-learning (F10)
Cognitive exertions that require their activities (F11)

Efficiency:
Quality around the audiovisual (F3)
Aesthetic and technical elements (F4)
Navigation and interaction (F6)
Mandatory and use of technology (F7)



Example1: Identifying Quality Space Dimensions

ID Requirement	ID Factor	% fulfilment of requirements in the ideal system	% fulfilment of requirements in the actual system
1	F1	10	68.3
2	F1	10	42.5
3	F2	10	42.5
4	F2	10	52.5
5	F2	10	49.17
6	F3	10	52.92
7	F4	10	67.5
8	F5	10	70.42
9	F5	10	72.08
10	F5	10	68.58
11	F5	10	72.5
12	F5	10	61.25
13	F6	10	59.17
14	F6	10	55
15	F6	10	55.83
16	F6	10	42.92
17	F7	10	54.28
18	F2	10	45.4
19	F8	10	60
20	F9	10	73.75
21	F9	10	66.67
22	F10	10	72.92
23	F11	10	63.33

Example1: Identifying Quality Space Dimensions

Dimension: Functionality (F1, F5)

DFuncionalidade = 1/20 * (10*85.83+10*68.33) + 1/50 * (10*86.67+10*84.58+10*85.42+10*87.5+10*73.75)

= 77.08 * 0.5 .5 * 0.5 = 80.33

Indica a Example1: Identifying Quality Space Dimensionsrelevância do factor na dimensão Funcionalidade

ANO LECTIVO: 2000/2001
C/RECURSO A X-TEC

Sistema Ideal
Peso Requisitos

	F/R	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20	R21	R22	R23	Total Peso
FACTORES	F1	10	10																						20
	F2			10	10	10													10						40
	F3						10																		10
	F4							10																	10
	F5								10	10	10	10	10												50
	F6													10	10	10	10								40
	F7																	10							10
	F8																			10					10
	F9																				10	10			20
	F10																						10		10
	F11																							10	10

Sistema Real
% Cumprimento Requisitos

	F/R	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20	R21	R22	R23	
FACTORES	F1	85,83	68,33																						F1
	F2			64,17	65,42	62,08													62,92						F2
	F3						69,17																		F3
	F4							84,17																	F4
	F5								86,67	84,58	85,42	87,5	73,75												F5
	F6													73,75	68,33	76,67	58,75								F6
	F7																	67,08							F7
	F8																			75					F8
	F9																				79,58	76,67			F9
	F10																						90		F10
	F11																							76,67	F11

Example1: Identifying Quality Space Dimensions

Dimensão: Adaptability (F2, F8, F9, F10, F11)

DAdaptability = 1/40 * (10*64.17+10*65.42+10*62.08+10*62.92) + 1/10 * (10*75) + 1/20 * (79.58 + 10 * 76.67) +
= 1/10 * (10*90) 1/10 * (10*76.67)

= 63.64 * 0.2 + 75 * 0.2 + 78 * 0.2 + 90 * 0.2 + 76.67 * 0.2 = 76.6 0.2

Indica a relevância do factor na dimensão Adaptabilidade

ANO LECTIVO: 2000/2001
C/RECURSO A X-TEC

Sistema Ideal
Peso Requisitos

FACTORES	F/R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total Peso
	F1	10	10																						20
	F2			10	10	10													10						40
	F3						10																		10
	F4							10																	10
	F5								10	10	10	10	10												50
	F6													10	10	10	10								40
	F7																	10							10
	F8																			10					10
	F9																				10	10			20
	F10																						10		10
	F11																							10	10

Sistema Real
% Cumprimento Requisitos

FACTORES	F/R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	
	F1	85,83	68,33																						F1
	F2			64,17	65,42	62,08													62,92						F2
	F3						69,17																		F3
	F4							84,17																	F4
	F5								86,67	84,58	85,42	87,5	73,75												F5
	F6													73,75	68,33	76,67	58,75								F6
	F7																	67,08							F7
	F8																			75					F8
	F9																				79,58	76,67			F9
	F10																						90		F10
	F11																							76,67	F11

Example1: Identifying Quality Space Dimensions

Dimension: Efficiency(F3, F4, F6, F7)

DEficiência = 1/10 * (10*69.17) + 1/10 * (10*84.17) + 1/40 * (10*73.75+10*68.33+10*76.67+10*58.75) + 1/10 * (10*67.08)

= 69.17 * 0.25 .17 * 0.25 + 69.38 * 0.25 + 90 * 0.25 + 67.08 * 0.25 = 72.45

ANO LECTIVO: 2000/2001
C/RECURSO A X-TEC

Sistema Ideal
Peso Requisitos

Indica a relevância do factor na dimensão Eficiência

FACTORES	REQUISITOS																							
F/R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total Peso
F1	10	10																						20
F2			10	10	10													10						40
F3						10																		10
F4							10																	10
F5								10	10	10	10	10												50
F6													10	10	10	10								40
F7																	10							10
F8																			10					10
F9																				10	10			20
F10																						10		10
F11																							10	10

Sistema Real
% Cumprimento Requisitos

FACTORES	REQUISITOS																							
F/R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	
F1	85,83	68,33																						F1
F2			64,17	65,42	62,08													62,92						F2
F3						69,17																		F3
F4							84,17																	F4
F5								86,67	84,58	85,42	87,5	73,75												F5
F6													73,75	68,33	76,67	58,75								F6
F7																	67,08							F7
F8																			75					F8
F9																				79,58	76,67			F9
F10																						90		F10
F11																							76,67	F11

Example1: Calculation of Deviation and System Quality

D = √ ((1-80.33/100) ^2 + (1-76.69/100) ^2 + (1-72.45/100) ^2)

D = 0.41

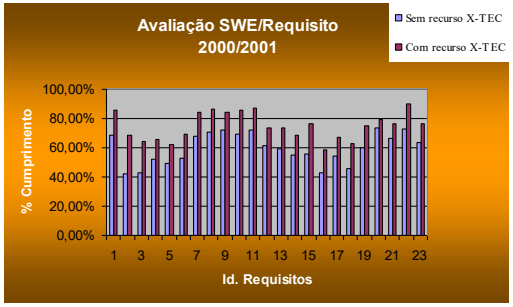
Q = 1- 0.41/√ (3)

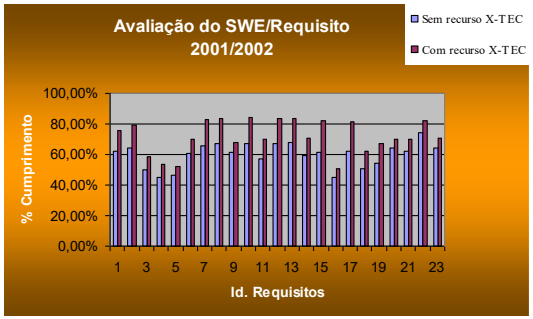
Q = 0,76

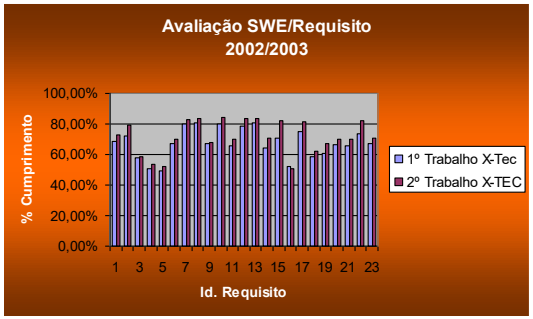
Q = 76 %

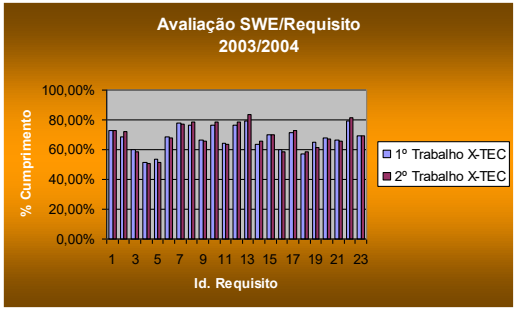
The system fulfills 76% of the purpose for which it was developed.

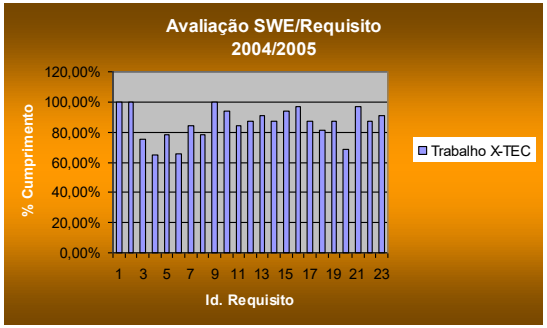
Example1: Calculation of Deviation and System Quality





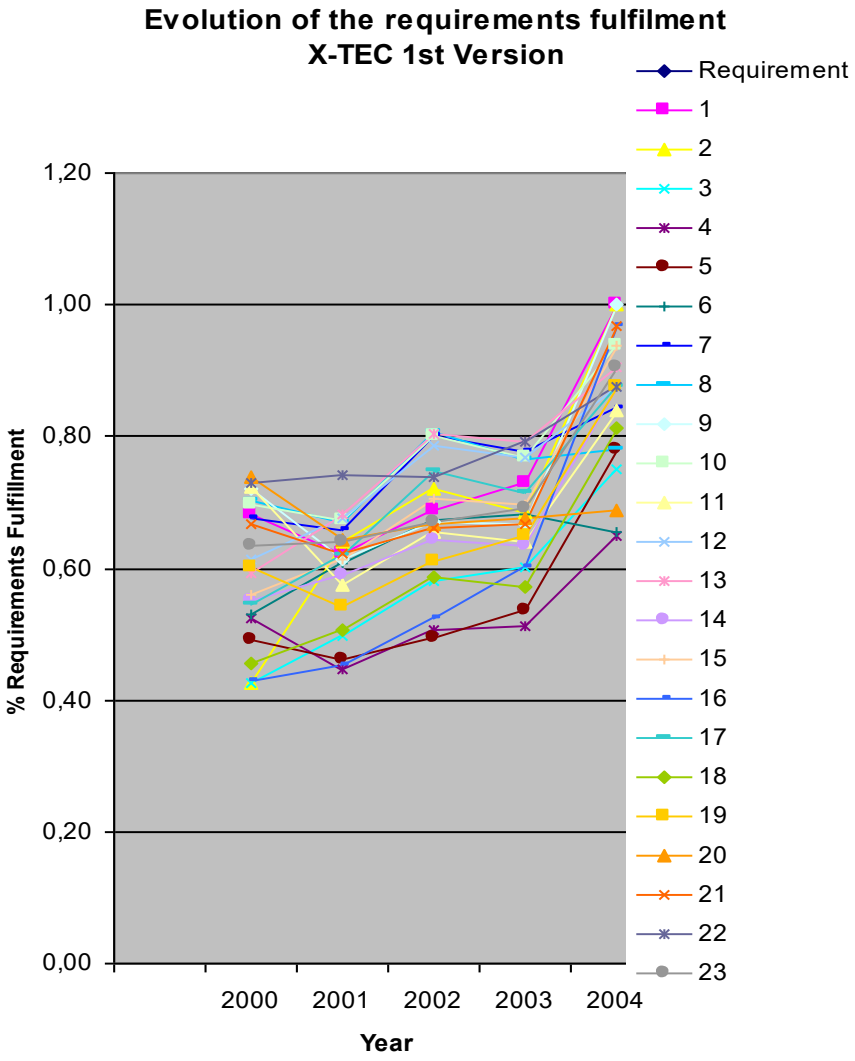




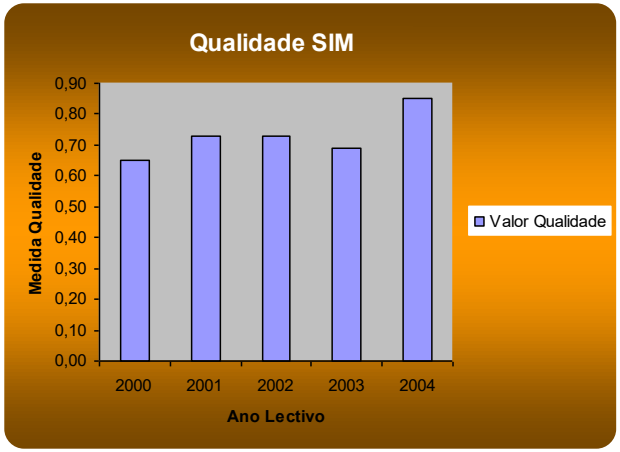


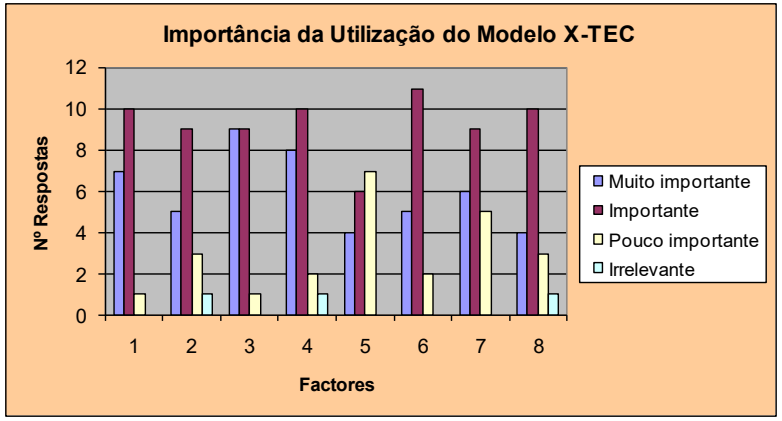
OThe system fulfills 76% of the purpose for which it was developed.s resultados experimentais permitem observar que o desempenho e, consequentemente, a qualidade dos sistemas produzidos pelos alunos na disciplina foi aumentado, chegando até, em alguns casos a atingir uma percentagem de cumprimento dos critérios de 100% em relação às especificações iniciais, identificadas pelos grupo disciplinar.

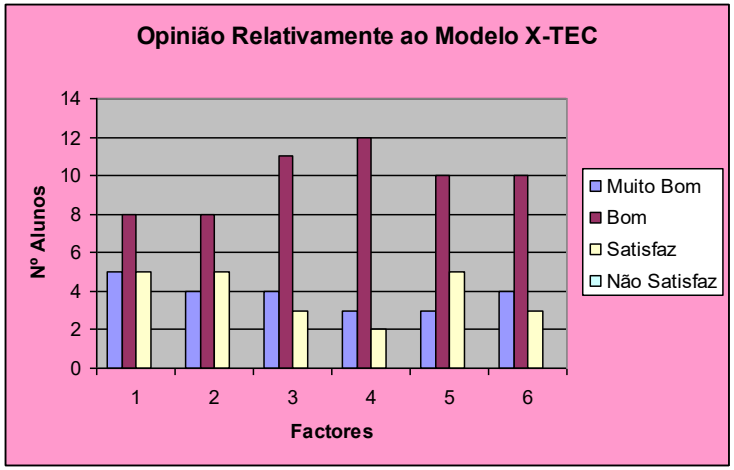
Analysis of the results on Multimedia System class



Example1: Calculation of Deviation and System Quality







Educational technology



E-book





Mobile Learning



Virtual Worlds



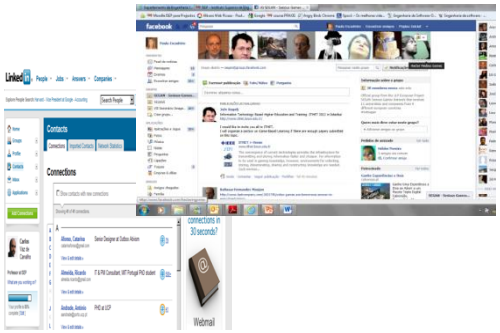
Serious Games



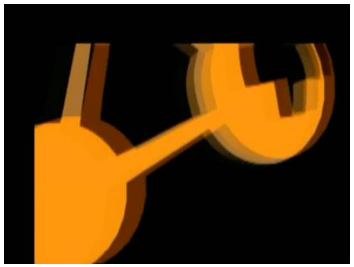
Virtual garden



Social Networks



Multimedia Medicine



Digital Interactive Stories



ISEP Channel - YouTube



Digital Interactive Toys

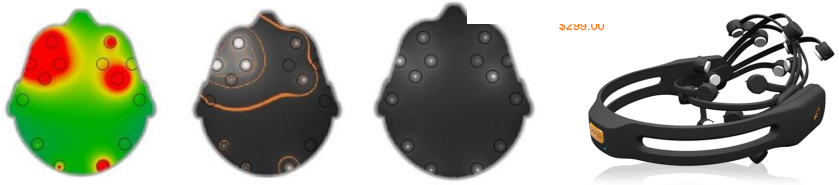


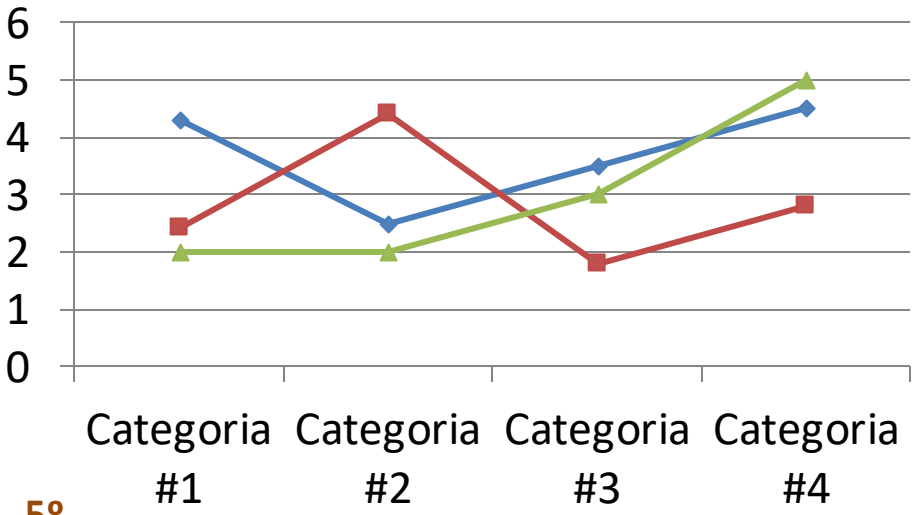
Haptic Systems



R&D

1º Learning Profile Identifier

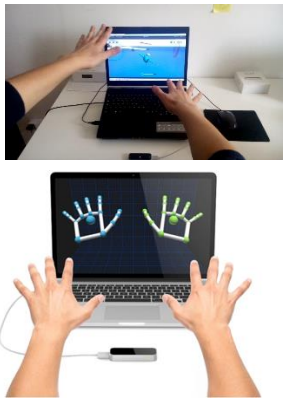




2º ACT – age (Automatic Cognitive Technologies for Active Ageing



3º Magic Hands



4º Virtual Sign

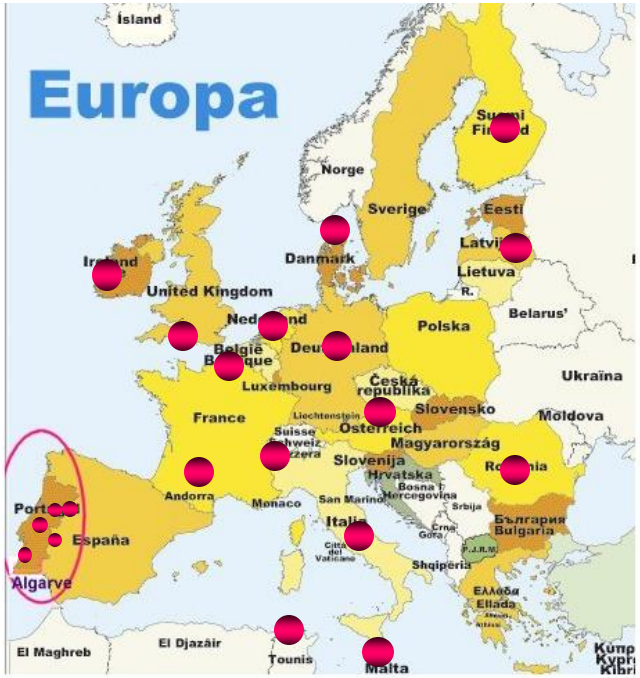


5º ACE - Assisted Communication for Education

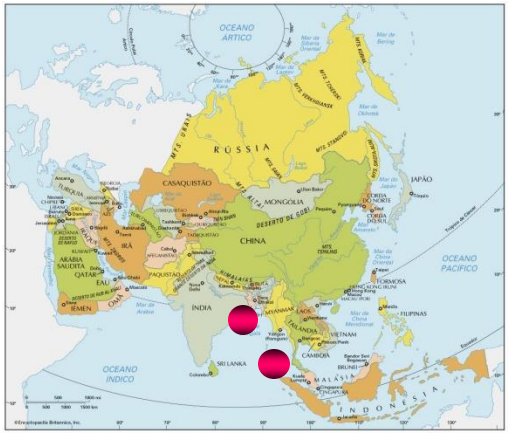


QEF

Dissemination







QEF

Thank you!

Paula Escudeiro